Practical NO.10
Name - Sakshi Vijaykumar Muthiyan
Roll No. -
Title -Data Visualization using matplotlib
Problem Statement: Analyzing Air Quality Index (AQI) Trends in a City
Dataset: "City_Air_Quality.csv"
Description: The dataset contains information about air quality measurements in a specific city over a period of time. It includes attributes such as date, time, pollutant levels (e.g., PM2.5, PM10, CO), and the Air Quality Index (AQI) values. The goal is to use the matplotlib library to create visualizations that effectively represent the AQI trends and patterns for different pollutants in the city.

In [10…
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
# settings file paths for both datasets on desktop
path_pollutants = r"C:\Users\muthi\OneDrive\Desktop\PRACTICAL EXAM\DMV\ar
path_aqi = r"C:\Users\muthi\OneDrive\Desktop\PRACTICAL EXAM\DMV\archive (
print("Loading original Pune air quality datasets...")
df_p = pd.read_csv(path_pollutants)
df_a = pd.read_csv(path_aqi)
# task 1 & 3: cleaning headers and merging datasets on common Date key co
df_p.columns = df_p.columns.str.strip()
df_a.columns = df_a.columns.str.strip()
# merging both datasets to get all pollutants and aqi metrics together
df_aqi = pd.merge(df_p, df_a[['Date', 'AQI']], on='Date', how='inner')
print("\n=== Task 2: Merged Dataset Matrix Shape ===")
print("Dimensions Data Shape:", df_aqi.shape)
df_aqi[['Date', 'PM2.5', 'PM10', 'CO', 'AQI']].head(3)
```

```
Loading original Pune air quality datasets...

=== Task 2: Merged Dataset Matrix Shape ===
Dimensions Data Shape: (8010, 10)
```

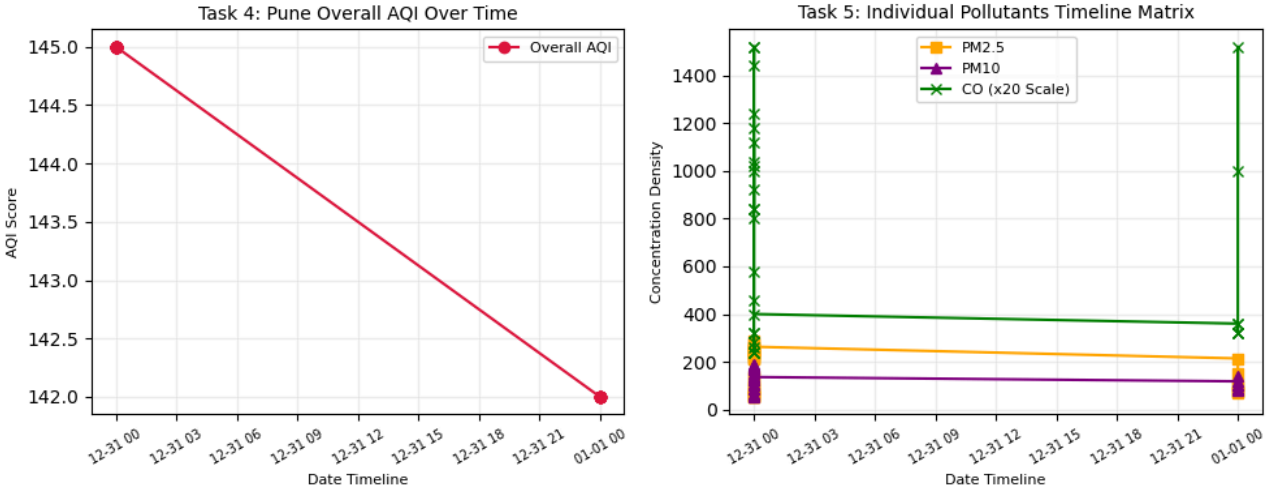
Out[10]:

	Date	PM2.5	PM10	CO	AQI
0	2023-12-31	211.0	191.0	59.0	145.0
1	2023-12-31	225.0	185.0	62.0	145.0
2	2023-12-31	240.0	182.0	50.0	145.0

In [17…
```
# task 3: explicitly identify columns relevant to plotting tasks
print("=== Task 3: Visualizing Variables Extracted ===")
print("X-Axis DataType Timeline Component:", df_aqi['Date'].dtype)
```

```
print("Y-Axis Target Pollutant Streams Array:", ['AQI', 'PM2.5', 'PM10',
# sorting rows sequence and slicing top 30 rows to prevent data overflow
df_sorted = df_aqi.sort_values('Date').head(30)
# Creating a 10x4 layout grid for side-by-side subplots configuration
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
# --- Left Chart: Task 4 Overall AQI Trend ---
axes[0].plot(df_sorted['Date'], df_sorted['AQI'], marker='o', color='crim
axes[0].set_title('Task 4: Pune Overall AQI Over Time', fontsize=10)
axes[0].set_xlabel('Date Timeline', fontsize=8)
axes[0].set_ylabel('AQI Score', fontsize=8)
axes[0].tick_params(axis='x', rotation=30, labelsize=7)
axes[0].grid(True, alpha=0.2)
axes[0].legend(fontsize=8)
# --- Right Chart: Task 5 Individual Pollutants Level Trend ---
axes[1].plot(df_sorted['Date'], df_sorted['PM2.5'], marker='s', color='or
axes[1].plot(df_sorted['Date'], df_sorted['PM10'], marker='^', color='pur
axes[1].plot(df_sorted['Date'], df_sorted['CO'] * 20, marker='x', color='
axes[1].set_title('Task 5: Individual Pollutants Timeline Matrix', fontsi
axes[1].set_xlabel('Date Timeline', fontsize=8)
axes[1].set_ylabel('Concentration Density', fontsize=8)
axes[1].tick_params(axis='x', rotation=30, labelsize=7)
axes[1].grid(True, alpha=0.2)
axes[1].legend(fontsize=8)
# layout adjustment block to trim blank padding margins away from pages
plt.tight_layout()
plt.show()
```

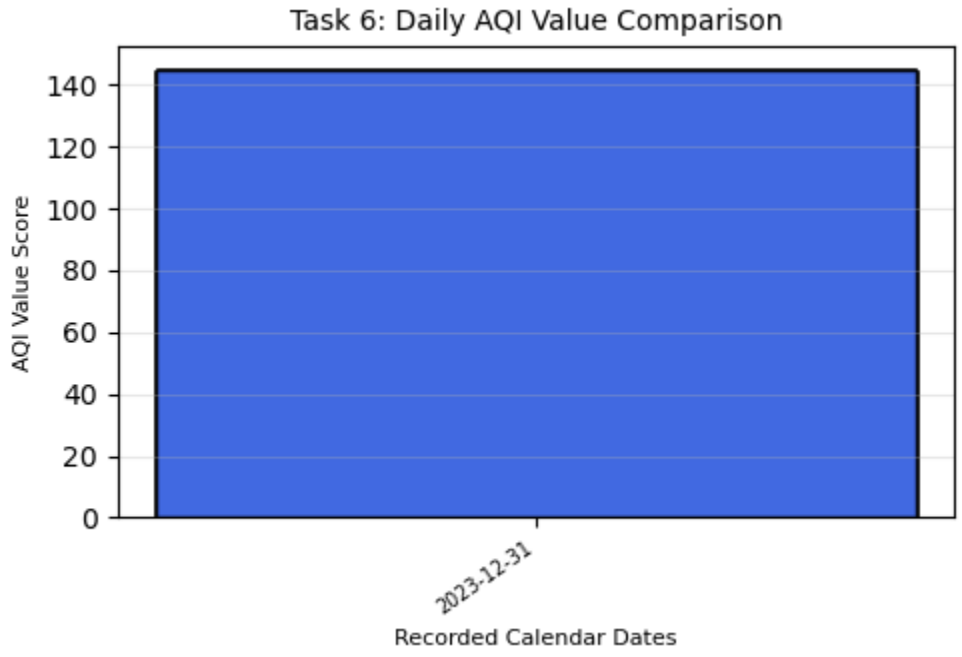
```
=== Task 3: Visualizing Variables Extracted ===
X-Axis DataType Timeline Component: datetime64[ns]
Y-Axis Target Pollutant Streams Array: ['AQI', 'PM2.5', 'PM10', 'CO']
```



In [18…

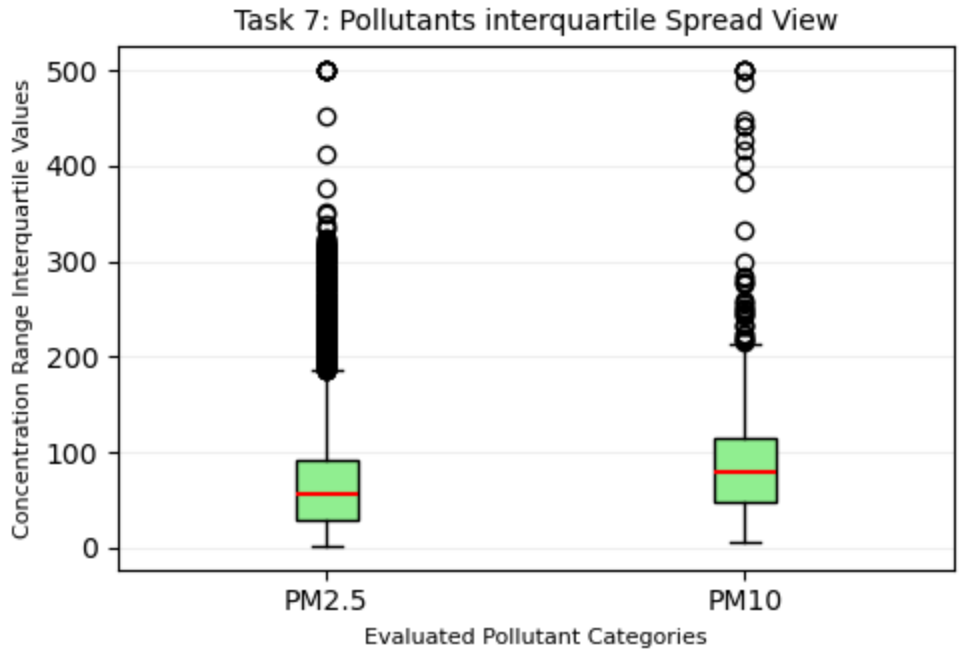
```
# task 6: vertical bar plot comparing values across different dates snaps
plt.figure(figsize=(5, 3.5))
bar_sample = df_sorted.head(10) # 10 records snapshot to keep graph bars
date_labels = bar_sample['Date'].dt.strftime('%Y-%m-%d')
plt.bar(date_labels, bar_sample['AQI'], color='royalblue', edgecolor='bla
plt.title('Task 6: Daily AQI Value Comparison', fontsize=10)
plt.xlabel('Recorded Calendar Dates', fontsize=8)
plt.ylabel('AQI Value Score', fontsize=8)
```

```
plt.xticks(rotation=35, ha='right', fontsize=7)
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



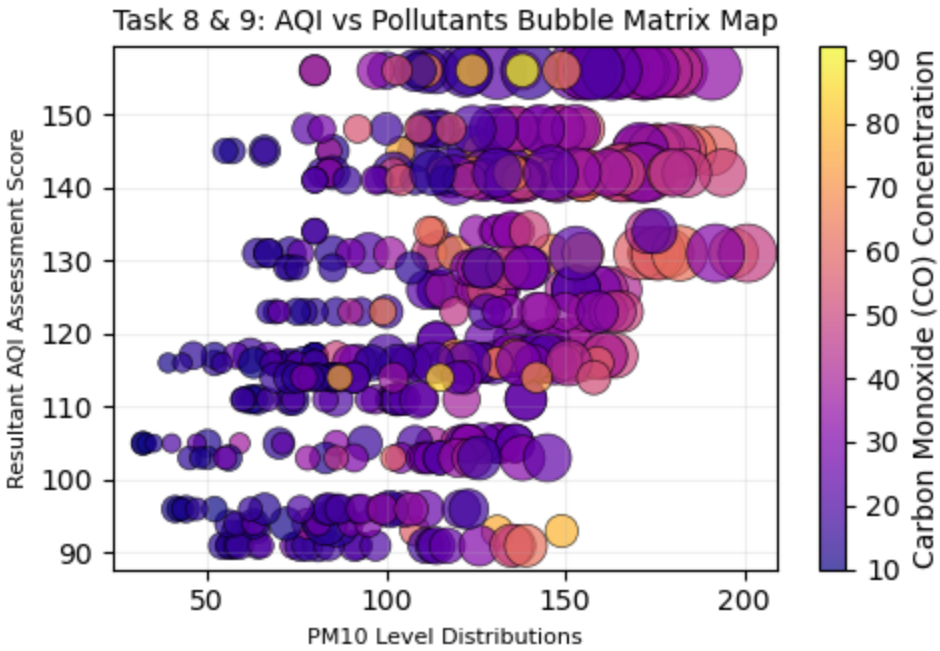
In [19…

```
# task 7: box plots to analyze pollutant metrics distributions density sp
plt.figure(figsize=(5, 3.5))
distribution_list = [df_aqi['PM2.5'], df_aqi['PM10']]
plt.boxplot(distribution_list, labels=['PM2.5', 'PM10'], patch_artist=Tru
            boxprops=dict(facecolor='lightgreen', color='black'),
            medianprops=dict(color='red', linewidth=1.5))
plt.title('Task 7: Pollutants interquartile Spread View', fontsize=10)
plt.xlabel('Evaluated Pollutant Categories', fontsize=8)
plt.ylabel('Concentration Range Interquartile Values', fontsize=8)
plt.grid(axis='y', alpha=0.2)
plt.tight_layout()
plt.show()
```



In [20…

```
# task 8 & 9: scatter bubble charts with full styling custom additions
plt.figure(figsize=(5, 3.5))
bubble_size = df_aqi['PM2.5'].head(500) * 1.5 # bubble scaling density li
scatter_plot = plt.scatter(df_aqi['PM10'].head(500), df_aqi['AQI'].head(5
                           cmap='plasma', alpha=0.7, edgecolor='black', li
# task 9 advanced labels and color schema decorations additions
plt.title('Task 8 & 9: AQI vs Pollutants Bubble Matrix Map', fontsize=10)
plt.xlabel('PM10 Level Distributions', fontsize=8)
plt.ylabel('Resultant AQI Assessment Score', fontsize=8)
plt.colorbar(scatter_plot, label='Carbon Monoxide (CO) Concentration')
plt.grid(True, alpha=0.2)
plt.tight_layout() # trims padding borders to minimize sheet printing cou
plt.show()
```


Task 8 & 9: AQI vs Pollutants Bubble Matrix Map
Resultant AQI Assessment Score
150
140
130
120
110
100
90
50
100
150
200
PM10 Level Distributions
Carbon Monoxide (CO) Concentration
90
80
70
60
50
40
30
20
10